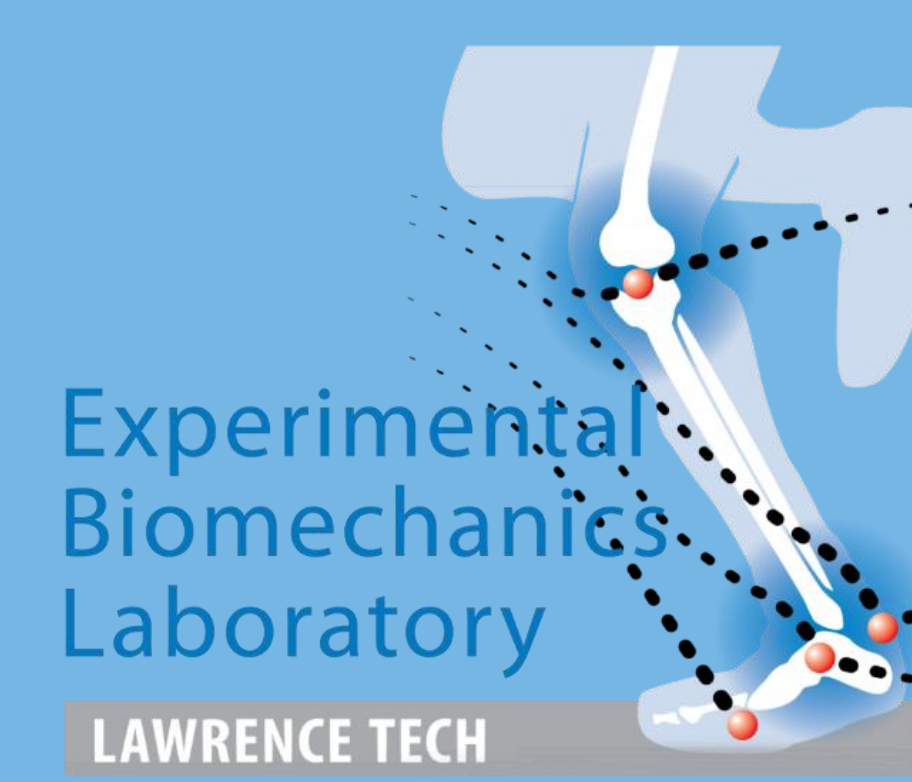


Valgus Bending of the Knee Reduces the Tibiofemoral Compression Required to Cause ACL Injury

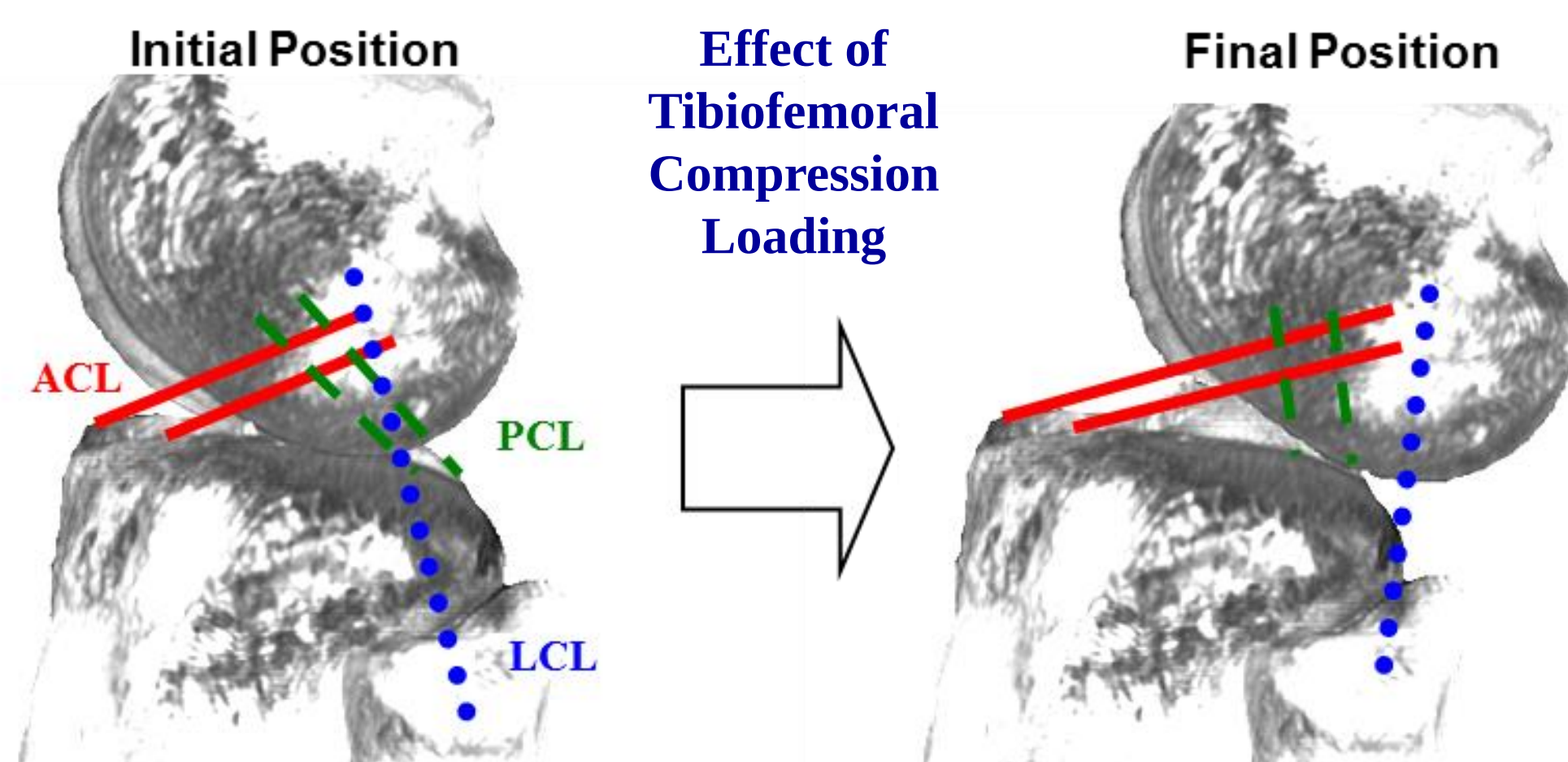


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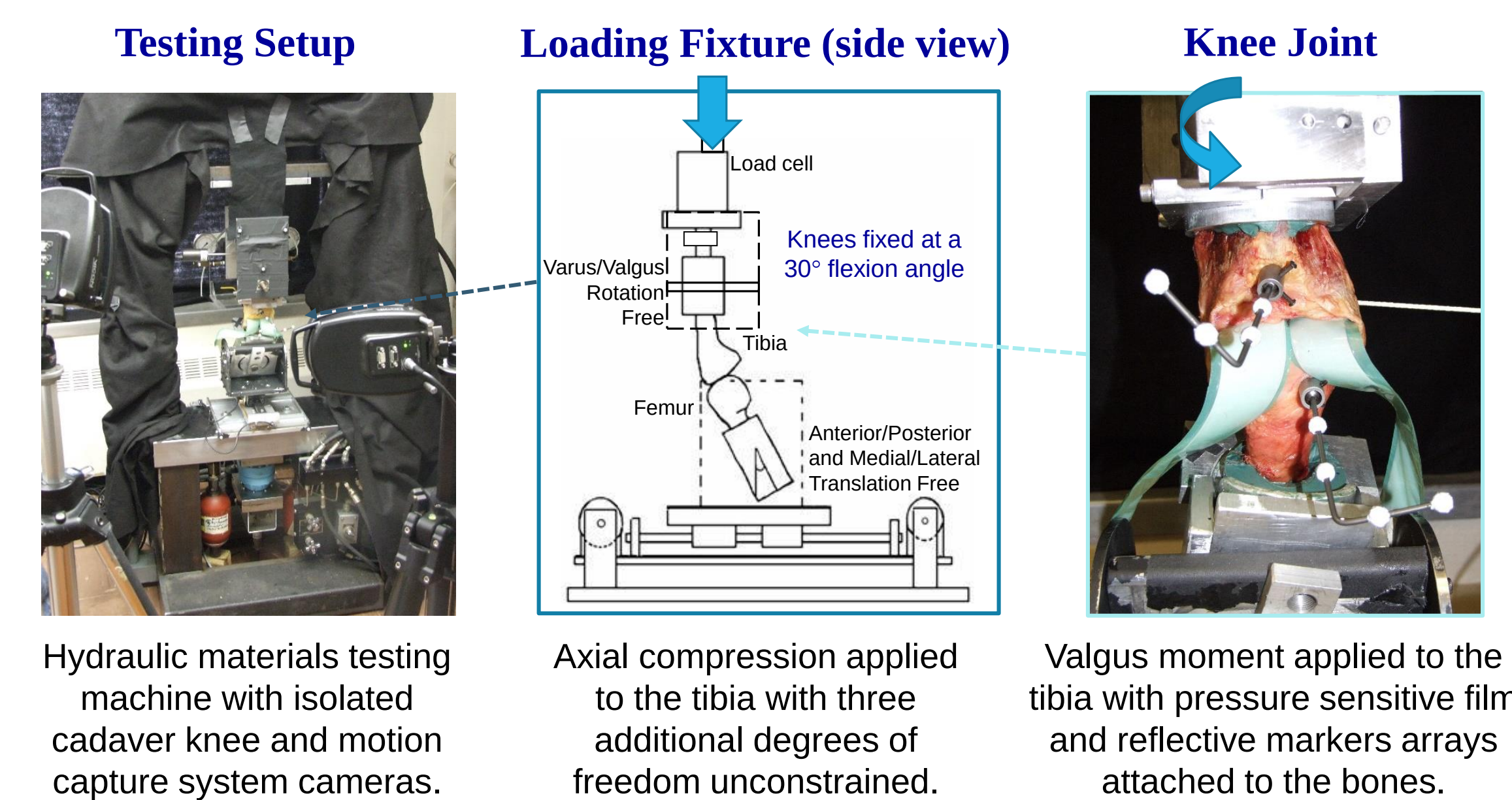
Introduction

- Anterior Cruciate Ligament (ACL) injuries are frequently caused by non-contact mechanisms that involve rapid deceleration, such as “Valgus Collapse”¹.
- The ground reaction force during a jump landing can be 4-6 xBW, and the knee reaction force can be up to 15 xBW². These forces are a combination of compression and shear based on the geometry of the tibiofemoral joint³.
- Excessive tibiofemoral compression (TFC) has been shown to rupture the ACL by producing anterior tibial subluxation⁴.
- The goal of this study was to induce ACL rupture in isolated knee joints by tibiofemoral compression (TFC) while simulating sports injuries like landing from a jump.



Methods

- Seven paired, isolated knee joints were potted in epoxy and loaded in a hydraulic materials testing system.
- Knees were positioned at 30° flexion with relative displacement and valgus bending unconstrained.
- TFC with or without a VB moment preload (10Nm) was applied randomly to limbs at repeated, increasing magnitudes until injury.

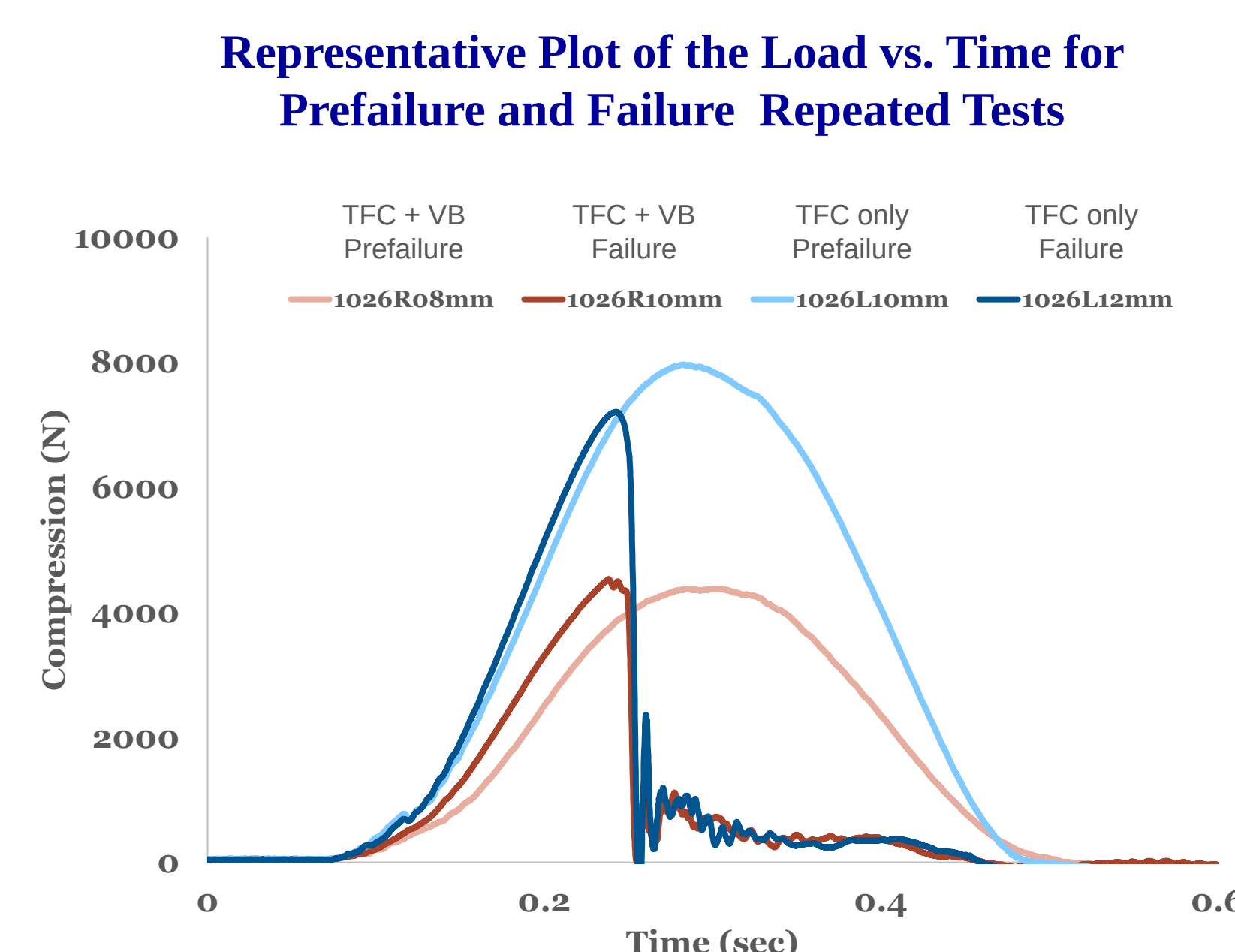


- Pressure-sensitive film recorded the ratio of TFC between the medial and lateral compartments.
- CT scans were analyzed to measure the slope of the overall tibial plateau and the posterior aspect of the plateau for each compartment.

Hypothesis

- When the knee is in a more vulnerable position, such as slightly flexed with small valgus bending (VB) moment, the TFC force that is required to cause isolated injury to the ACL will be reduced.**

Results

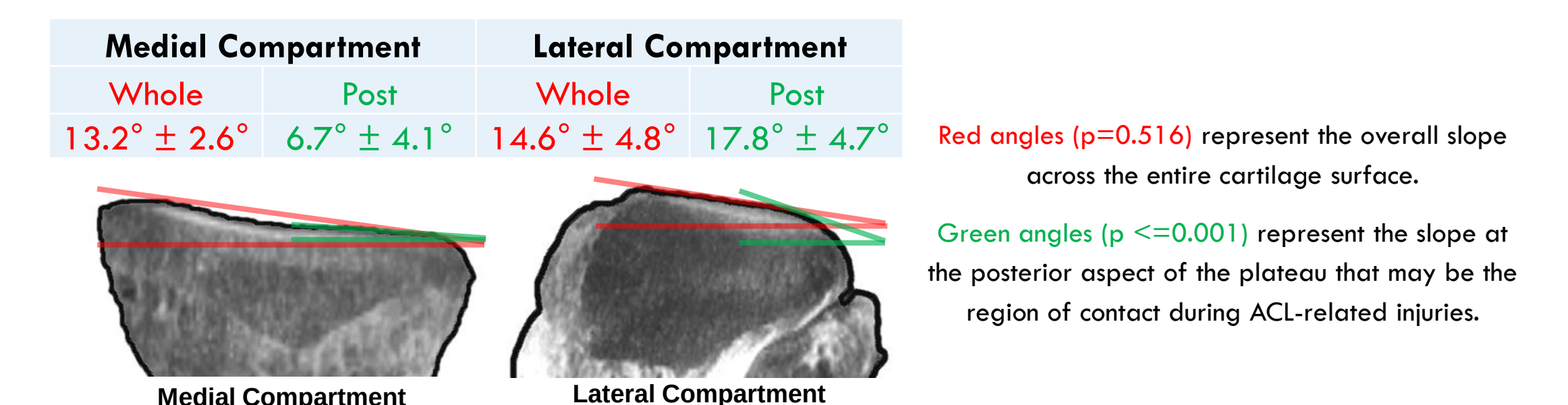


Results

- The injuries included tibial avulsion of the ACL (n=5), partial ACL tear (n=4), ACL midsubstance tear (n=3), femoral avulsion of ACL (n=1), and tibial fracture (n=1).
- ACL injuries in combined loading limbs occurred at a significantly lower compressive force compared with those in TFC alone limbs ($3.7 \pm 1.3\text{kN}$ vs. $6.0 \pm 1.5\text{kN}$, $p=0.03$).

Specimen	Leg	TFC (N)	VB (Nm)	Injury Mode	Leg	TFC (N)	VB (Nm)	Injury Mode
1026	Left	7200	10	Partial ACL tear	Right	4500	0	Partial ACL tear
1180	Left	5400	10	Tibia avulsion of ACL	Right	8200	0	Tibia avulsion of ACL
1065	Left	4500	10	Femur avulsion of ACL	Right	2961	0	Partial ACL tear
1477	Left	4235	10	Tibia fracture	Right	6200	0	ACL tear
1441	Left	3202	10	Tibia avulsion of ACL	Right	4721	0	Partial ACL tear
1466	Left	4200	10	ACL tear	Right	2150	0	ACL tear
33608	Right	1509	10	Tibia avulsion of ACL	Left	6250	0	Tibia avulsion of ACL

- The posterior aspect of the lateral tibial plateau had a significantly higher slope than the medial plateau.



- For TFC-alone knees the contact was similar between medial and lateral compartments. For TFC+VB knees the contact was shifted almost entirely to the lateral compartment.

Discussion

- Tibia plateau geometry is a factor that can increase athletes' risk for ACL injuries⁵.
- Many “Valgus Collapse” ACL injuries involve significant TFC loading⁶.
- VB may reduce the TFC required to cause ACL-related injuries due to the inherent instability of the lateral compartment.**

References: (1) Olsen et al., 2004 Am J Sports Med 32:1002-1012 (2) Meyer & Haut, 2005 J Biomech 38:2311-2316 (3) Meyer & Haut, 2008 J Biomech 41:3377-3383 (4) Hewett et al., 1996 Am J Sports Med 24:765-773 (5) Hashemi et al., 2008 J Bone Jt Surg 90:2724-2734 (6) Koga et al., 2010 Am J Sports Med 38:2218-2225